

HUBBARD BROOK ECOSYSTEM STUDY

SAMPLE NOTES

PRECIPITATION SAMPLES

(TUES)
DATE COLLECTED: 08 MAR 2020 Initials: TW

Conversion Factors

10 in. Bulk Rain Funnel: NET divided by 51.1 = catch

N/A if Overfilled

12 in. Bulk Snow Bucket: NET divided by 69.3 = catch

SITE	TIME (EST)	COLLECTOR VOLUME (ml)	CATCH (mm)	ETI CATCH (mm)	Coll. EFF. (% of ETI)
RG-22	<u>0840</u>	<u>1467</u>	<u>21.2</u>		
RG-1	<u>1205</u>	<u>1479</u>	<u>21.3</u>		
RG-4	<u>1135</u>	<u>1459</u>	<u>21.1</u>		
RG-23	<u>1005</u>	<u>1697</u>	<u>24.5</u>		

Volumes Collected				
	RG-22	RG-1	RG-4	RG-23
Gross g	<u>2428</u>	<u>2439</u>	<u>2419</u>	<u>2658</u>
Tare g	<u>961</u>	<u>960</u>	<u>960</u>	<u>961</u>
Net g	<u>1467</u>	<u>1479</u>	<u>1459</u>	<u>1697</u>

SITE	CONTAMINATION (0 = none 1 = low -----> 5 = high)						dBASE FIELD code?	SAMPLE REJECTED @LEVEL =	Database CODES checked ?
	Fine PM	Coarse PM	Bird FECES	Insects	Color	GB Ring			
RG-22	<u>4</u>	<u>3+ seeds</u> <u>Back</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>2</u>			
RG-1	<u>3</u>	<u>3</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>2</u>			
RG-4	<u>3+</u>	<u>2</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>2</u>			
RG-23	<u>4</u>	<u>2 seeds</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>2</u>			

Precipitation amount
1.12 inches / 28.4 mm
ETI collector

ADDITIONAL COMMENTS: Precip was snow and rain, buckets had rim ice.

SG-STA/22 = 11" SG-1 = 5" SG-4 = 15", SG-23 = 22" FS Snow data: 21-41 cm Depth
Gusty winds/snow showers during collection 7-24 cm SWE

Applicable dBASE FIELD codes:
(See SOP Manual)

970 = Obvious Contamination: Sample REJECTED

969 = Possible Contamination: Check Return Analyses

955 = QNS: Quantity Not Sufficient for analyses (< 100 mL NET catch)

precip-fieldnotes.xls 2017 v RG-4

HUBBARD BROOK ECOSYSTEM STUDY

SAMPLE NOTES

Streamwater SamplesDate Collected: 08 MAR 2022 ^(TUES) Initials: TW

SITE	TIME (EST)	TEMP (°C)	GAGE HT.	HYDROGR L/R/F/PK	COLOR (0 = none:1 = low---->5 = high)	SEDIMENT	REMARKS	dBASE FIELD code?
WS-1	1210	0.9°	.433	F	1	0	snow/ice bridges	
WS-2	1155	0.8°	.406	F	0	1	"	
WS-3	1145	1.0°	.578	F	1	1	mostly open w/ s/i bridges	
WS-4 (Dup)	1115	0.8°	.517	F	0	1	Algae	
WS-5	1050	0.4°	.544	F	0	0	snow/ice bridges	
WS-6	1100	0.6°	.422	F	1	1	"	
WS-7	1025	0.3°	.743	F	1	0	ic/sc opw sample hole	
WS-8	1015	0.1°	.684	F	1-	0	ice/snow bridges	
WS-9	0950	0.1°	.772	F	3	1	Locked up - DOC foam weir but shifted ice	
Hubbard BK	1230	0.1°		R	2	0	ice shifting	
Mlake Outlet	1310	1.2°		RL	1	0	OTB opw Staff	

ADDITIONAL COMMENTS: Rain/melt created channel openings, ice shifting - some opw w/ snow bridges. Not a blowout. Falling hydrographs. Lots of windblown debris, limbs. Some open ground around tree wells, boulders, S. Facing steep slopes/knobs.

Applicable dBASE FIELD codes: 911= High Flow Event in Progress; 912 = Drought Conditions (syringe from Standing Pool), 969 = possible problems (note)

955 = QNS: Quantity Not Sufficient for analyses

master sample notes 2017

See page 5 for ANC's

HUBBARD BROOK ECOSYSTEM STUDY

METROHM / 3 STAR LAB ANALYSES

DATE RUN: 09 MAR 2022 (3*) RUN By: TW
Meter CALIBRATION: See Calibration form for Standard Procedure
Conductivity Correction Factor = 1.12x at 18.8 °C Lab Temp

Samples:	DATE COLLECTED	Metrohm pH	3 Star pH	SpCond @ 25°C	Remarks Codes
<u>Precipitation</u>	<u>08 MAR 2022</u>				
STATION	<u>220308</u>		<u>5.03</u>	<u>9.0</u>	
RG-1			<u>5.10</u>	<u>8.8</u>	
RG-4			<u>5.11</u>	<u>8.7</u>	
RG-23	<u>220308</u>		<u>5.21</u>	<u>7.4</u>	

		Metrohm pH	Metrohm ANC	ANC end pt. pH	3 Star pH	SpCond @ 25°C	Remarks Codes
<u>Streamwater</u>							
WS-1	<u>220308</u>				<u>5.44</u>	<u>14.6</u>	
WS-2					<u>5.51</u>	<u>8.2</u>	
WS-3					<u>5.61</u>	<u>9.6</u>	
WS-4					<u>5.93</u>	<u>9.2</u>	
WS-5					<u>5.49</u>	<u>9.7</u>	
WS-6					<u>5.55</u>	<u>10.0</u>	
WS-7					<u>5.73</u>	<u>9.5</u>	
WS-8					<u>5.34</u>	<u>10.5</u>	
WS-9					<u>4.52</u>	<u>17.7</u>	
Hubbard BK					<u>6.05</u>	<u>10.3</u>	
Mlake Outlet	<u>220308</u>				<u>6.23</u>	<u>31.7</u>	

ADDITIONAL COMMENTS: Metrohm computer down. Samples refrigerated for later run.

HBEF Stream and M.Lake Routine Sample DIC Datadate: **10MAR2022**

Samples collected 03/08/2022

Aqueous STANDARDS

type	DIC	peak ht	adjusted	ATT -X
	set =	mm	mm	
DIW blank	0	10	0	16
	0	10	0	16
	0	12	0	16
100 µM	100	31	21	16
	100	31	21	16
	100	31	19	16
200 µM	200	54	44	16
	200	54	44	16
	200	52	40	16
400 µM	400	103	93	16
	400	101	91	16
	400	99	87	16

Statistics

Values from matrix at M5 -->

slope = 4.389
 slope ± 0.083
 Y-int = 6.8
 Y-int ± 4.2
 R-sq = 0.9964

Laboratory Conditions

Lab Temp = 18.8 °C
 Barom Pressure = 770 mm Hg
 Analyst's Initials: TW 48%RH
 NaCO3 Standards
 0.1M Stock Dated: 3/10/22
 0.01M Working Solution Dated: 3/10/22

Samples

	inj 1	inj 2	inj 3	ATT -X	Avg Peak	±SD Peak	Samples	DIC	Calc
	1	2	3					µM/L	err±
ML-70	48	48	47	16	47.7	0.6	ML-70	209	3
	0	0	0	16	0.0	0.0	0	0	0
	0	0	0	16	0.0	0.0	0	0	0
	0	0	0	16	0.0	0.0	0	0	0
	0	0	0	16	0.0	0.0	0	0	0
W-1	20	17	20	16	19.0	1.7	W-1	83	8
W-2	14	14	15	16	14.3	0.6	W-2	63	3
W-3	12	12	12	16	12.0	0.0	W-3	53	0
W-4	14	13	13	16	13.3	0.6	W-4	59	3
W-5	10	10	10	16	10.0	0.0	W-5	44	0
W-6	13	13	13	16	13.0	0.0	W-6	57	0
W-7	11	10	11	16	10.7	0.6	W-7	47	3
W-8	9	9	10	16	9.3	0.6	W-8	41	3
W-9	12	11	10	16	11.0	1.0	W-9	48	4
HBk	13	12	12	16	12.3	0.6	HBk	54	3
	0	0	0	16	0.0	8.0	0		
							0		
Lab Air	16	16	16	16	16.0	0.0	Lab Air	70	0
Outside Air	11	11	13	16	11.7	1.2	Outside Air	51	5
Gas 1000	24	25	24	16	24.3	0.6	Gas 1000	107	3
Gas 3000	73	73	75	16	73.7	1.2	Gas 3000	323	5

NOTES

120 mA current

std=83.3

std=246.3

HUBBARD BROOK ECOSYSTEM STUDY

METROHM / 3 STAR LAB ANALYSES

DATE RUN:

24 MAR 2022
09 MAR 2022 (3*)

RUN By:

TW

Meter CALIBRATION:

See Calibration form for Standard Procedure

Conductivity Correction Factor = 1.12x

at: 18.8 °C Lab Temp

Samples:

DATE
COLLECTEDMetrohm
pH3 Star
pHSpCond
@ 25°CRemarks
CodesPrecipitation 08 MAR 2022

STATION	DATE COLLECTED	Metrohm pH	3 Star pH	SpCond @ 25°C	Remarks Codes
RG-1	220308	NA	5.03	9.0	
RG-4		No data	5.10	8.8	
RG-23	220308	NA	5.11	8.7	
			5.21	7.4	

Run
3/9/22StreamwaterMetrohm
pHMetrohm
ANCANC
end pt. pH (3/9) pH 3/243 Star
pH
SpCond
@ 25°CRemarks
Codes

STATION	DATE COLLECTED	Metrohm pH	Metrohm ANC	ANC end pt. pH (3/9) pH 3/24	3 Star pH	SpCond @ 25°C	Remarks Codes
WS-1	220308	5.48	+4.2	5.21 (5.44)	5.45	14.6	
WS-2		5.57	+0.3	5.41 (5.51)	5.53	8.2	
WS-3		5.68	+5.8	5.21 (5.61)	5.64	9.6	
WS-4		5.95	+15.1	5.18 (5.93)	5.98	9.2	
WS-5		5.48	+0.2	4.70 (5.49)	5.48	9.7	
WS-6		5.61	+7.5	5.46 (5.55)	5.58	10.0	
WS-7		5.80	+11.8	5.33 (5.73)	5.77	9.5	
WS-8		5.37	+0.2	4.95 (5.34)	5.36	10.5	
WS-9		4.51	-25.6	4.30 (4.52)	4.53	17.7	
Hubbard BK		6.07	+21.4	5.32 (6.05)	6.07	10.3	
Mlake Outlet	220308	6.35	+72.4	5.20 (6.23)	6.31	31.7	

ADDITIONAL COMMENTS:

Metrohm down, samples refrigerated for later run. 3* pH initial run in parentheses.

Resumed Titrations 3/24/22. Could not run Metrohm pH on precip. Buckets were dumped/washed for following weeks catch. Snowmelt chemistry

Applicable codes: 911= High Flow Event in Progress;

955= QNS: Quantity Not Sufficient for analyses(<100 mL NET catch)

912 = Drought Conditions (syringe from Standing Pool); 969 = possible problems (note)